

IDR305 21H Introduction to Economics for Sport Management Exam

Instructions (read carefully)

- a. **3 hours and 30 minutes, including all practical and technical actions needed to upload in Inspira.**
- b. **All supporting materials permitted. It is not allowed to cooperate with or receive help from others.**
- c. **Justify all your answers with economic concepts, logical reasoning, and love for sport.**

Good Luck!

Question 1 (20 points)

Suppose there are two independent countries Northland and Southland. Those countries consume two goods, oil (X) and wine (Y).

Assume that in Northland there are 150 workers and each of them can produce 5 units of oil or 3 units of wine. Assume that in Southland there are 200 workers and each of them can produce 2 units of oil or 2 units of wine.

- a. Please draw a production possibilities frontier of each of these countries. **(5 points)**
- b. Each country wants to produce and consume 250 units of oil to keep a healthy life. Given that we are in a closed economy (i.e., no trades are allowed), how many units of wine will be consumed by each country? **(4 points)**
- c. Now assume that the two countries are allowed to trade with the rest of the world, such that 5 units of oil is worth 4 units of wine. Please draw a production possibilities frontier for each of the two countries. **(4 points)**
- d. If each country still wants to produce 250 units of oil, what will be the consumption of wine of each of these countries? **(4 points)**
- e. Please explain whether trade is profitable or not for each of these countries. **(3 points)**

Question 2 (25 points).

You are given the following functions:

- 1) Demand function: $Q^D = 65 - 5p$.
- 2) Supply function: $Q^S = 8p$

Where p is the price of a good in US dollars (\$) and Q^D and Q^S are the quantity demanded and supplied, respectively for that specific good.

- a. What would be the demanded quantity if the price is 7 \$? **(2 points)**
- b. What would be the demanded quantity if the price is 3 \$? **(2 points)**
- c. What would be the supplied quantity if the price is 1 \$? **(2 points)**

- d. What would be the supplied quantity if the price is 4 \$? **(2 points)**
- e. What would be the price and the quantity in the equilibrium? Show a graph with the equilibrium point, the consumer, and producer surplus. **(7 points)**
- f. Which is the producer surplus? **(4 points)**
- g. Which is consumer surplus? **(4 points)**
- h. Which is the total social welfare? **(2 points)**

Question 3 (20 points)

You are given the following firm's production function:

TP (Q)	L	MP	FC	VC	TC	MC	ATC	AVC
0	0				90			
1	1							
2	5							
3	10							
4	16							
5	24							
6	34							
7	50							

One unit of labor (L) costs 5\$

- a. Complete the table. **(8 points)**
- b. What is the minimal price that the firm will produce in the short term? **(2 points)**
- c. What is the minimal price that the firm will produce in the long term? **(2 points)**
- d. It is known that each unit of production can be sold in the market for 40\$ ($p=40$). What would be the production of the firm? **(2 points)**
- e. Would the firm produce in the short term? **(3 points)**
- f. Would the firm produce in the long term? **(3 points)**

Question 4 (15 points)

There is a market with 9 workers who should be allocated between two fields, given the following production function:

Field A

L	TP
0	0
1	10
2	18
3	24
4	28
5	30

Field B

L	TP
0	0
1	12
2	19
3	24
4	27
5	28

- How many workers would work in each field if it is known that the price of one produced unit in field **A** is **5\$** and the price of one produced unit in field B is **4\$**? **(7 points)**
- What would be the salary of each worker? **(3 points)**
- What would be the profit of the owners of each field? **(5 points)**

Question 5 (5 points)

Snow FC, a regional football team in Lapland, is planning to buy a new player: HotWater. The team wants him to sign a 5 years contract and it is willing to spend 1500€ for the player. The value of the player (the money spent by the team to transfer him) will increase by 10% each year if he scores more than 10 goals per year. However, if he scores less than 10 goals per year his value will increase by only 5%. Which will be the future value of the player at the end of the contract if he is going to score 4 and 7 goals in the first two seasons, respectively, and 11, 14, and 12 goals in the following ones?

Question 6 (5 points)

Suppose you have an elastic good (Molde FK jerseys). What would happen to the total revenue of the supplier (Molde FK storekeeper) if it increases the price of the good?

Question 7 (10 points)

Suppose the following market with two groups of consumers (A and B), where:

- $Q_A^D = 11 - 4p$, where Q_A^D is the quantity demanded of a good and p is the price of a good in USD by each consumer in group A
- $Q_B^D = 9 - 6p$, where Q_B^D is the quantity demanded of a good and p is the price of a good in USD by each consumer in group B

Suppose that group A has 5 consumers while group B has 3 consumers.

- What is the aggregate demand for group A? **(3 points)**
- What is the aggregate demand for group B? **(3 points)**
- What is the aggregate demand of the whole market if the market price is 2.5 USD? What is the aggregate demand if the price is 1 USD? (*Hint: pay*

attention for which value of p the demanded quantity is zero for group A and group B.) **(4 points)**